

Obstacle-Aware Quadrotor Control via NMPC and Artificial Potential Fields

Gizem Doğa Filiz, İsmail Cem Yılmaz

Abstract—This paper presents a quaternion-based Nonlinear Model Predictive Control (NMPC) framework for quadrotor obstacle avoidance, and studies *where* obstacle avoidance should be performed: as formal collision-avoidance constraints *inside* the NMPC optimal control problem, or by a separate Artificial Potential Field (APF) planner that generates references the NMPC merely tracks. The controller uses a 13-dimensional rigid-body state with quaternion attitude, avoiding Euler-angle singularities near the tilt threshold at which both architectures fail, and is solved online with an acados SQP-RTI solver at a 1 s horizon and 50 ms sampling rate; obstacles are detected online from a 2-D lidar via DBSCAN clustering. Evaluated in Gazebo on a five-obstacle in-line slalom at a matched ≈ 0.9 m clearance, the in-NMPC keep-out constraint navigates reliably up to 1.5 m/s while the APF-reference plus NMPC-tracker reaches 2.2 m/s; both fail at higher speed through the same attitude-runaway instability, and the APF advantage is shown to stem from its wider look-ahead rather than from the architecture itself. The average solver time is 8.6 ms, well within the 50 ms control budget.

I. INTRODUCTION

Autonomous quadrotor navigation in cluttered environments requires real-time trajectory planning and precise low-level control. Classical approaches decouple these tasks: a high-level planner generates a geometric path, while a separate controller tracks it. Artificial Potential Fields (APF) [1] provide a computationally lightweight mechanism for reactive path planning, yet they lack inherent guarantees on dynamic feasibility. Nonlinear Model Predictive Control (NMPC) [3], in contrast, explicitly enforces vehicle dynamics and actuation limits over a receding horizon, but requires an accurate reference trajectory. Within NMPC, collision avoidance can be embedded directly as soft or hard constraints or as control barrier functions [5]; recent agile-flight work has further compared NMPC against differential-flatness tracking [6].

These two mechanisms can both prevent collisions, but they place the avoidance logic in different parts of the stack. This study therefore poses a concrete question: *should obstacle avoidance be enforced as formal collision-avoidance constraints inside the NMPC, or delegated to an APF planner whose collision-free references the NMPC only tracks?* Both are implemented on the same quaternion NMPC and the same lidar perception, and compared on an in-line slalom course at a deliberately matched obstacle clearance, so that only the architecture differs.

In the NMPC papers we surveyed for this work, quadrotor studies split into two patterns: those with a full-quaternion state target agile tracking or time-optimal flight without in-loop obstacle constraints (e.g. [7], [14], [6], [8]), while

those including collision-avoidance constraints often adopt Euler or cascaded attitude representations (e.g. [9], [10]). Our formulation combines the two: a full-quaternion NMPC state together with in-loop keep-out obstacle constraints. This combination matters here because the speed ceiling identified in Section IV is set by an attitude-runaway instability, a coupling between fast lateral avoidance and body tilt that only becomes visible when the OCP predicts the full attitude dynamics: reducing the attitude to Euler angles introduces a singularity near the tilt threshold where the runaway occurs, and deferring attitude control to a cascaded inner loop hides the coupling from the outer OCP altogether. The regime where this coupling becomes visible is one where the tilt required for lateral avoidance approaches the vehicle’s angular limits, either at high cruise or with clearances that force large swerves; this is the regime we sweep in Section IV.

The main contributions are:

- A quaternion-based NMPC formulation, solved with acados SQP-RTI, with both a landing-cone soft constraint and online soft *keep-out* collision-avoidance constraints embedded in the optimal control problem.
- A reactive APF horizon builder that feeds per-step references to the NMPC, enabling online avoidance without pre-planned waypoints, with lidar-based obstacle detection via DBSCAN clustering providing the obstacles to both modes.
- A controlled comparison of in-NMPC keep-out avoidance against APF-reference tracking at matched clearance, showing that the APF speed advantage stems from its wider look-ahead, not the architecture.
- Identification of the shared speed ceiling of both architectures as an attitude-runaway instability at high tilt, made visible by the full-quaternion NMPC formulation.

The remainder of the paper is organised as follows. Section II presents the methodology, covering the dynamic model, NMPC formulation, APF planner, and lidar perception. The experimental setup is described in Section III. Results are presented and discussed in Section IV. Finally, Section V concludes the paper.

II. METHODOLOGY

A. Quadrotor Dynamic Model

1) *State and input definitions:* The quadrotor is modelled as a rigid body with mass m and diagonal inertia $\mathbf{J} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ (numerical values in Table I). The system state is

$$\mathbf{x} = [\mathbf{p}^\top \mathbf{v}^\top \mathbf{q}^\top \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{13}, \quad (1)$$

where $\mathbf{p} \in \mathbb{R}^3$ is the world-frame position, $\mathbf{v} \in \mathbb{R}^3$ the world-frame linear velocity, $\mathbf{q} = [q_w, q_x, q_y, q_z]^\top \in \mathbb{S}^3$ the unit quaternion, and $\boldsymbol{\omega} = [p, q, r]^\top \in \mathbb{R}^3$ the body-frame angular velocity, whose components p, q, r are the roll, pitch and yaw rates about the body x, y, z axes respectively. The control input is the generalised wrench

$$\mathbf{u} = [f_{\text{tot}} \ \tau_x \ \tau_y \ \tau_z]^\top \in \mathbb{R}^4. \quad (2)$$

We use the generalised wrench $\mathbf{u}$ as the OCP input rather than the four rotor squared speeds Ω^2 directly. Both parametrisations describe the same feasible set through the invertible allocation $\mathbf{u} = \mathbf{B}\Omega^2$, so the choice does not restrict modelling capability; we prefer the wrench because the rigid-body dynamics are natural in that space, the tracking-cost weights on $\mathbf{u}$ correspond to physically meaningful axes (thrust and roll/pitch/yaw torque effort), and the platform's actuator limits translate into a simple box constraint on each wrench component (Table I). The accompanying per-rotor feasibility guarantee is stated with the OCP constraints in Section II-B.3.

2) *Equations of motion:* Newton's second law in the world frame gives $\dot{\mathbf{p}} = \mathbf{v}$ and

$$\dot{\mathbf{v}} = \frac{f_{\text{tot}}}{m} \mathbf{R}(\mathbf{q}) \mathbf{e}_3 - g \mathbf{e}_3, \quad (3)$$

where $\mathbf{R}(\mathbf{q}) \in SO(3)$ is the rotation matrix built from the unit quaternion $\mathbf{q}$ (with $SO(3)$ the group of 3×3 orthogonal matrices with unit determinant, i.e. the space of rigid-body rotations), and $g = 9.81 \text{ m/s}^2$. Expanding the quaternion-rotation product:

$$\dot{v}_x = \frac{2(q_w q_y + q_x q_z)}{m} f_{\text{tot}}, \quad (4)$$

$$\dot{v}_y = \frac{2(q_y q_z - q_w q_x)}{m} f_{\text{tot}}, \quad (5)$$

$$\dot{v}_z = \frac{1 - 2q_x^2 - 2q_y^2}{m} f_{\text{tot}} - g. \quad (6)$$

Quaternion kinematics are governed by

$$\dot{\mathbf{q}} = \frac{1}{2} \boldsymbol{\Xi}(\mathbf{q}) \boldsymbol{\omega}, \quad \boldsymbol{\Xi}(\mathbf{q}) = \begin{bmatrix} -q_x & -q_y & -q_z \\ q_w & -q_z & q_y \\ q_z & q_w & -q_x \\ -q_y & q_x & q_w \end{bmatrix}. \quad (7)$$

The rotational dynamics follow Euler's rigid-body equation:

$$\dot{p} = [\tau_x - (I_{zz} - I_{yy})qr] / I_{xx}, \quad (8)$$

$$\dot{q} = [\tau_y - (I_{xx} - I_{zz})pr] / I_{yy}, \quad (9)$$

$$\dot{r} = [\tau_z - (I_{yy} - I_{xx})pq] / I_{zz}. \quad (10)$$

The full model is written compactly as $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$.

3) *Motor allocation:* Individual squared rotor speeds $\Omega^2 \in \mathbb{R}^4$ are obtained by inverting the “+”-configuration allocation matrix $\mathbf{u} = \mathbf{B}\Omega^2$, where

$$\mathbf{B} = \begin{bmatrix} k_f & k_f & k_f & k_f \\ 0 & \ell k_f & 0 & -\ell k_f \\ -\ell k_f & 0 & \ell k_f & 0 \\ k_m & -k_m & k_m & -k_m \end{bmatrix}, \quad (11)$$

with thrust and moment coefficients k_f, k_m and arm length ℓ (Table I).

B. Nonlinear Model Predictive Control

1) *Optimal control problem:* At each sampling instant t_k the NMPC solves the finite-horizon OCP

$$\min_{\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + \ell_N(\mathbf{x}_N) \quad (12)$$

subject to

$$\mathbf{x}_{k+1} = F(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \quad (13)$$

$$\mathbf{x}_0 = \hat{\mathbf{x}}(t_k), \quad (14)$$

$$\mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, N-1, \quad (15)$$

$$h(\mathbf{x}_k) \geq 0, \quad k = 0, \dots, N, \quad (16)$$

where $F(\cdot)$ is the discrete-time model obtained by integrating $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ with a four-stage explicit Runge-Kutta (ERK4) scheme, and $\hat{\mathbf{x}}(t_k)$ is the current state estimate. The horizon parameters N and T_s are given in Table I.

2) *Cost function:* A nonlinear least-squares cost structure is adopted with output map $\mathbf{h}(\mathbf{x}, \mathbf{u}) = [\mathbf{x}^\top, \mathbf{u}^\top]^\top \in \mathbb{R}^{17}$:

$$\ell(\mathbf{x}_k, \mathbf{u}_k) = \|\mathbf{h}(\mathbf{x}_k, \mathbf{u}_k) - \mathbf{y}_k^{\text{ref}}\|_{\mathbf{W}}^2, \quad (17)$$

where $\mathbf{y}_k^{\text{ref}} = [\mathbf{x}_k^{\text{ref}\top}, \mathbf{u}_{\text{hov}}^\top]^\top$ with $\mathbf{u}_{\text{hov}} = [mg, 0, 0, 0]^\top$. The weight matrix $\mathbf{W} = \text{diag}(\mathbf{Q}, \mathbf{R})$ is

$$\mathbf{Q} = \text{diag}(\underbrace{5, 5, 5}_{\mathbf{p}}, \underbrace{3, 3, 3}_{\mathbf{v}}, \underbrace{1.5, 1.5, 1.5, 1.5}_{\mathbf{q}}, \underbrace{25, 25, 6}_{\boldsymbol{\omega}}), \quad (18)$$

$$\mathbf{R} = \text{diag}(0.01, 0.10, 0.10, 0.20). \quad (19)$$

The reduced yaw-rate weight $Q_r = 6 \ll Q_{p,q} = 25$ reflects the lower yaw authority relative to roll/pitch. The terminal cost is

$$\ell_N(\mathbf{x}_N) = \|\mathbf{x}_N - \mathbf{x}_N^{\text{ref}}\|_{\mathbf{W}_N}^2, \quad \mathbf{W}_N = \text{diag}(P_s \mathbf{Q}_{\mathbf{p}, \mathbf{v}, \mathbf{q}}, \mathbf{0}_3), \quad (20)$$

with $P_s = 5$ and zero terminal penalty on angular rates.

Attitude cost formulation: For attitude tracking we penalise the component-wise squared difference between the state quaternion $\mathbf{q} = (q_w, q_x, q_y, q_z)$ and the reference quaternion $\mathbf{q}_{\text{ref}}$, with an antipodal sign fix $\mathbf{q} \leftarrow -\mathbf{q}$ whenever $q_w \cdot q_{\text{ref},w} < 0$ to remove the double-cover ambiguity. This is not the geodesic error on $SO(3)$; the geometrically correct attitude error is $2 \cdot \arccos(|(\mathbf{q} \otimes \mathbf{q}_{\text{ref}}^{-1})_w|)$. The component-wise form is chosen for two reasons. First, because $\mathbf{h}(\mathbf{x}, \mathbf{u})$ contains the quaternion components directly, its Jacobian $\mathbf{J} = \partial \mathbf{h} / \partial [\mathbf{x}, \mathbf{u}]$ is constant and block-sparse, so the Gauss-Newton Hessian $\mathbf{J}^\top \mathbf{W} \mathbf{J}$ used inside SQP-RTI does not need to be recomputed at every iteration and the QP structure that HPIPM exploits is preserved. Second, around the reference the component-wise and the geodesic cost differ only by a constant scaling factor and are locally equivalent up to a $\sin(\theta/2)/(\theta/2)$ factor. The empirical impact of this approximation on our tracking runs is quantified in Section IV.

3) *Constraints:* Four constraints are imposed on every stage of the horizon: two *hard* input constraints on the actuation (actuator box bounds and per-rotor feasibility), and two *soft* state constraints (landing cone and obstacle keep-out). They are described in turn below.

a) *Actuator box bounds.*: The feasible set $\mathcal{U}$ enforces actuator limits component-wise (Table I). The tight yaw-torque limit reflects the physical limit $f_{\text{hov}}k_m/k_f \approx 0.19\text{ Nm}$ imposed by the motor mixer; the conservative bound eliminates yaw-saturation oscillations observed during validation.

b) *Per-rotor feasibility.*: Bounding each component of $\mathbf{u}$ (2) alone is not sufficient to guarantee that a commanded wrench can actually be produced by the four rotors: at box corners the resulting wrench can map, through $\mathbf{B}^{-1}$, to a negative squared rotor speed for at least one motor, a state fixed-pitch propellers cannot occupy. Per-rotor feasibility is therefore added to the OCP as a *hard* constraint

$$\Omega^2 = \mathbf{B}^{-1}\mathbf{u}_k \geq \mathbf{0}, \quad k = 0, \dots, N-1, \quad (21)$$

where Ω^2 denotes the four squared rotor speeds recovered from $\mathbf{u}$ through the inverse mixer. The notation reflects the aerodynamic thrust-to-speed law $T_i = k_f \Omega_i^2$, which makes the allocation $\mathbf{u} = \mathbf{B}\Omega^2$ linear in Ω^2 . HPIPM enforces (21) exactly at every stage with no nonlinear iteration, so the OCP itself only outputs wrenches the four rotors can actually produce; the mixer clip in the low-level driver is retained only as a safety net against residual floating-point drift.

Two conventions are common in quadrotor NMPC. The first uses a high-level abstraction as its input, typically collective thrust together with either body-rate or attitude commands, with rotor allocation deferred to a low-level controller (e.g. [9], [14], [8]). The second uses the four per-rotor thrusts $T_i = k_f \Omega_i^2$ directly as decision variables with box bounds on each T_i (e.g. [7]). Our formulation sits between the two: it keeps the wrench parametrisation of the first family (for the cost-weight and box-bound reasons discussed above) and adds the per-rotor feasibility guarantee of the second through (21). The rotor speed Ω_i itself is not adopted as a decision variable, since the thrust law is quadratic in Ω_i but linear in Ω_i^2 ; parametrisation on Ω_i^2 (equivalently on T_i) is what keeps the allocation and (21) linear.

Only the lower bound is enforced inside the OCP. The physical maximum rotor speed is enforced post hoc by the same mixer step: in the tracking regime studied here the observed Ω^2 stays more than an order of magnitude below its physical cap (peak $\Omega_i^2 \approx 1.4 \times 10^4$ against $\Omega_{\text{max}}^2 \approx 1.4 \times 10^6$), and adding a non-binding finite upper bound was found empirically to degrade HPIPM's SQP-RTI warm-start during low-altitude descent without any tracking benefit.

c) *Landing cone.*: To prevent hard landings a soft state constraint is imposed at every stage:

$$h_{\text{land}}(\mathbf{x}_k) = v_z + \alpha p_z \geq 0, \quad \alpha = 2.0 \text{ s}^{-1}. \quad (22)$$

At $p_z = 0.3\text{ m}$ the maximum permissible descent rate is 0.6 m/s . The constraint is softened via a lower-bound slack $s_k \geq 0$, adding $W_{\text{land}}s_k^2$ to the stage cost with $W_{\text{land}} = 500$.

d) *Obstacle keep-out.*: When obstacle avoidance is enforced *inside* the NMPC, a soft keep-out constraint is imposed at every stage for each obstacle:

$$h_{\text{obs},i}(\mathbf{x}_k) = \|\mathbf{p}_k - \mathbf{p}_{o,i}\|^2 - (r_i + R_d)^2 \geq 0, \quad (23)$$

where $\mathbf{p}_{o,i}$ and r_i are the centre and radius of obstacle i and R_d is the drone radius (Table I). The constraint represents each obstacle by a single centre–radius pair; extensions to more elaborate geometries are discussed in Section IV (*Scope and outlook*). The obstacle centres and radii are online parameters, refreshed each control step from the lidar. Each constraint is softened with a lower-bound slack penalised by $W_{\text{obs}} = 10^4$, so a feasible control is always returned even under transient perception errors. The *same* NMPC acts as a pure tracker when no obstacles are supplied (APF-reference mode); activating (23) makes the controller itself responsible for clearance (in-NMPC mode). Because the keep-out acts on the NMPC's dynamically-feasible predicted trajectory, it constrains the path the vehicle actually flies; an APF reference, by contrast, is only kinematically collision-free and may be tracked with error, so the executed path can pass closer than the reference intends.

4) *SQP real-time iteration solver*: The OCP (12) is solved with the *Real-Time Iteration* (RTI) variant of Sequential Quadratic Programming as implemented in acados [3]. RTI performs a single SQP iteration per sampling interval: the nonlinear dynamics are linearised around the current iterate, the resulting convex QP is solved with HPIPM [4] using partial condensing, and the solution is immediately applied to the plant. The Hessian is approximated via Gauss–Newton, $\nabla^2 \mathcal{L} \approx \mathbf{J}_h^T \mathbf{W} \mathbf{J}_h$, guaranteeing positive semi-definiteness. The solver configuration and all key parameters are summarised in Table I.

C. APF-Based Obstacle Avoidance

1) *Potential field formulation*: Following [1], the workspace is modelled as a superposition of an attractive potential pulling toward the goal and repulsive potentials centred on obstacles. The corresponding force at position $\mathbf{p} \in \mathbb{R}^2$ is $\mathbf{F} = \mathbf{F}_{\text{att}} + \mathbf{F}_{\text{rep}}$, where

$$\mathbf{F}_{\text{att}} = k_{\text{att}}(\mathbf{p}_g - \mathbf{p}), \quad \|\mathbf{F}_{\text{att}}\| \leq k_{\text{att}}, \quad (24)$$

and, for each obstacle i with centre $\mathbf{p}_{o,i}$, radius r_i , and influence distance d_0 ,

$$\mathbf{F}_{\text{rep},i} = k_{\text{rep}} \left(\frac{1}{\rho_i} - \frac{1}{d_0} \right) \frac{1}{\rho_i^2} (\hat{\mathbf{r}}_i + \frac{1}{2} \hat{\boldsymbol{\tau}}_i), \quad \rho_i < d_0, \quad (25)$$

with $\rho_i = \|\mathbf{p} - \mathbf{p}_{o,i}\| - r_i - R_d$ the effective margin, $\hat{\mathbf{r}}_i$ the unit radial direction, and $\hat{\boldsymbol{\tau}}_i$ a signed tangent component. The tangent sign is precomputed from the cross product of the start-to-goal direction and the obstacle offset vector, ensuring consistent obstacle avoidance direction. This precomputed sign makes the vehicle deviate to the same side of each obstacle throughout the run, which is sufficient for the static in-line slalom studied here but can be limiting when obstacle configurations or goals change online; alternatives that we did not exercise here include randomising the sign per obstacle, replanning it from the current obstacle configuration at each control step, or learning a side-picking heuristic from data. The APF parameters k_{att} , k_{rep} , d_0 , and R_d are listed in Table I.

2) *Reactive horizon builder*: At each control step the function $\mathbf{F}(\mathbf{p})$ is evaluated at the current position and integrated forward for $N + 1$ steps of duration T_s to build the NMPC reference horizon:

$$\mathbf{p}_{k+1} = \mathbf{p}_k + T_s \cdot v_{\max} \frac{\mathbf{F}(\mathbf{p}_k)}{\|\mathbf{F}(\mathbf{p}_k)\|} \cdot \min\left(1, \frac{d_k}{2}\right), \quad (26)$$

where d_k is the distance to goal and $v_{\max}$ is the maximum speed (Table I). No pre-planned path is required; the system responds online to changes in obstacle configuration.

This builder is used in two regimes: a wide influence distance d_0 (smooth, early avoidance) and a tight one matched to the in-NMPC clearance. The in-NMPC mode instead uses a goal-only reference and delegates avoidance entirely to the keep-out constraint (23).

D. Lidar-Based Perception

The obstacle avoidance pipeline is driven by a 2-D lidar sensor mounted on the quadrotor. At each control step, the lidar point cloud is segmented using DBSCAN clustering [11] with parameters ε and $n_{\min}$ given in Table I. Here ε is the neighbourhood radius used to grow a cluster (two lidar points closer than ε are considered part of the same obstacle), and $n_{\min}$ is the minimum number of points a candidate cluster must contain before it is accepted (smaller groups are rejected as sensor noise). Each cluster is fitted with a bounding circle to estimate the obstacle centre $\mathbf{p}_{o,i}$ and radius r_i , which are supplied to the active avoidance module: the NMPC keep-out constraint (23) in in-NMPC mode, or the APF force computation in APF-reference mode. The perception module runs synchronously with the NMPC at 20 Hz, adding negligible latency (< 1 ms).

III. EXPERIMENTAL VALIDATION

A. Simulation Environment

All experiments were conducted in Gazebo using the tk3lab Docker environment. The simulated quadrotor is equipped with a 2-D lidar sensor for obstacle detection. The NMPC runs at 20 Hz on a standard desktop CPU; the perception module (DBSCAN clustering) runs synchronously at the same rate. All system and controller parameters are listed in Table I.

B. Experimental Procedure

The evaluation scenario is an **in-line slalom**: five cylinders of radius 0.4 m placed on the straight start–goal line, navigated in four configurations (pure-NMPC (tight), pure-NMPC (wide), APF+NMPC (tight), APF+NMPC (wide)). For each, the commanded speed is swept and the maximum reliable speed is recorded over repeated runs; obstacles are detected online via lidar and DBSCAN. The lidar sensor is configured with a maximum range of 6 m, so at each control step only obstacles within this look-ahead window are seen; the perception module therefore sets the horizon over which either avoidance mechanism can react.

TABLE I: System and Controller Parameters

Category	Parameter	Value
Vehicle	Mass m	1.280 kg
	Inertia (I_{xx}, I_{yy}, I_{zz})	$(22.9, 22.9, 22.1) \times 10^{-3}$ kg m ²
	Arm length ℓ	0.23 m
	Thrust coeff. k_f	6.5×10^{-4} N/(rad/s) ²
	Moment coeff. k_m	10^{-5} N m/(rad/s) ²
NMPC	Horizon N / T_s	20 / 0.05 s
	NLP solver	SQP-RTI (acados)
	QP solver	HPIPM (partial cond.)
	Integrator	ERK4
	Thrust bounds	$[0.4, 2.5]$ mg
	Torque bounds	$ \tau_{x,y} \leq 0.25, \tau_z \leq 0.06$ Nm
	Keep-out $R_d/W_{\text{obs}}/n_{\text{obs}}$	0.35 m / $10^4 / 5$
APF	$k_{\text{att}} / k_{\text{rep}}$	1.0 / 0.8
	d_0 (wide / tight)	2.5 / 0.3 m
	R_d (wide / tight)	0.50 / 0.20 m
	Cruise $v_{\max}$	1.5–2.8 m/s (swept)
Perception	Lidar max range	6.0 m
	DBSCAN ε	0.25 m
	Min. cluster $n_{\min}$	3 points

IV. RESULTS AND DISCUSSION

A. Obstacle Avoidance: In-NMPC Constraint vs. APF Reference

To isolate the effect of *where* avoidance is performed, all runs use the same in-line slalom: five cylinders of radius 0.4 m placed on the straight start–goal line at (3.5, 0.3), (6.5, −0.3), (9.5, 0.3), (12.5, −0.3) and (15.5, 0.3) m along an 18 m corridor. Because the obstacles sit on the line, a goal-directed reference points straight through them, so any clearance must be produced by the avoidance mechanism itself. Obstacles are detected online by lidar and DBSCAN.

1) *Pure-NMPC: avoidance inside the controller*: The *pure-NMPC* configuration uses a goal-only reference together with the keep-out constraint (23), so avoidance lives entirely in the controller. It is evaluated in two configurations that differ only in the size of the keep-out radius:

- **pure-NMPC (tight)**: the keep-out uses the nominal drone radius $R_d = 0.35$ m, giving a minimum obstacle clearance of ≈ 0.9 m matched to APF+NMPC (tight).
- **pure-NMPC (wide)**: the detected obstacle radius is inflated by an additional safety margin so that the effective keep-out radius matches APF+NMPC (wide), yielding a minimum clearance of ≈ 2.3 m.

Fig. 1 details a representative tight-clearance run: the lateral motion is produced entirely by the keep-out constraint, which holds the clearance to every obstacle above R_d while thrust and torque stay within their bounds. Panel (c) confirms the total thrust stays inside $[f_{\min}, f_{\max}]$ throughout the flight, and panel (d) shows the body torques never leave their box bounds. Panel (e) reports the per-obstacle clearance profile: each of the five cylinders is passed at ≈ 0.9 m at closest approach, matching the target margin. The wide-clearance configuration exists to test whether extra headroom in the keep-out is by itself enough to recover the speed of APF+NMPC (wide); it is reported below alongside the look-ahead analysis.

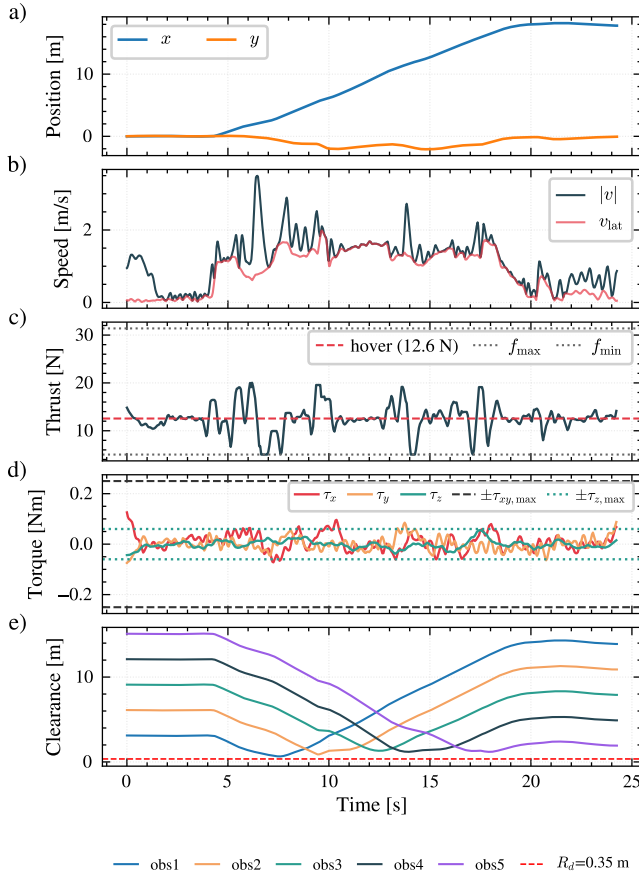


Fig. 1: Pure-NMPC slalom flight (avoidance in the keep-out constraint). (a) Position. (b) Total and lateral speed. (c) Total thrust with the NMPC input bounds $f_{\min} = 0.4 mg$ and $f_{\max} = 2.5 mg$ shown as dotted lines (hover $mg = 12.6 N$ dashed). (d) Body-frame torques τ_x, τ_y, τ_z with their input bounds $\pm\tau_{xy, \max} = \pm 0.25 Nm$ and $\pm\tau_{z, \max} = \pm 0.06 Nm$ as dotted lines. (e) Per-obstacle clearance ($R_d = 0.35 m$, dashed).

2) *Comparison with APF-reference tracking:* Pure-NMPC is compared against two configurations that instead delegate avoidance to an APF planner tracked by the same NMPC:

- **APF+NMPC (tight):** the APF influence and radius reduced ($d_0 = 0.3 m$, $R_d = 0.20 m$) to match the $\approx 0.9 m$ clearance of pure-NMPC.
- **APF+NMPC (wide):** the APF at its design point ($d_0 = 2.5 m$, $R_d = 0.50 m$), keeping a much larger $\approx 2.3 m$ clearance.

Fig. 2(a) shows the resulting paths for the tight-clearance comparison: the matched-clearance runs (pure-NMPC (tight), APF+NMPC (tight)) thread within $\approx 1 m$ of the obstacles, whereas the wide planner keeps much more clearance. The same figure also shows pure-NMPC (wide) in light blue, whose behaviour is discussed in the look-ahead analysis below. The path *shape* also differs by where avoidance is decided. The APF feeds the resultant potential-field vector at the current position as a velocity reference, so the reference already curves away from an obstacle while it is still far,

and the tracked path bends gradually and stays smooth. Pure-NMPC instead carries a goal-only reference that points straight through the obstacle; the lateral motion appears only when the predicted trajectory reaches the keep-out boundary (23), producing a later, sharper swerve that is visibly less smooth than the APF paths.

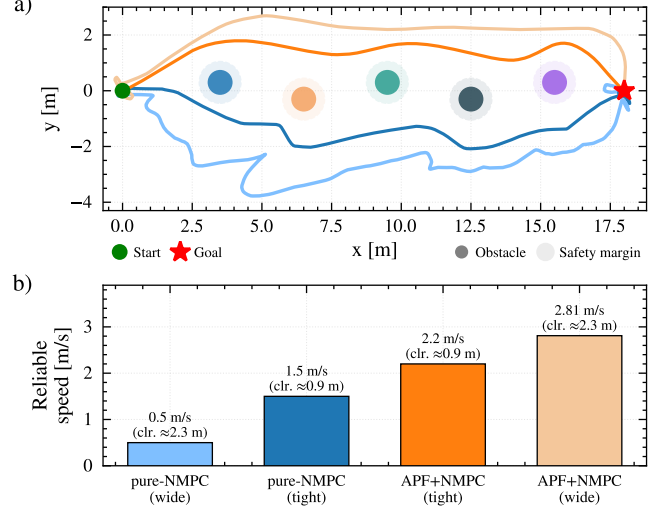


Fig. 2: Avoidance comparison (colours match across panels). (a) Paths for all four configurations at their respective clearances: pure-NMPC (tight) in dark blue, pure-NMPC (wide) in light blue, APF+NMPC (tight) in orange, APF+NMPC (wide) in peach; filled discs are obstacles, light rings the safety margin. (b) Maximum reliable speed at matched clearances (bars sorted by speed, with the clearance annotated above each bar).

Maximum reliable speed. The commanded speed was swept and the highest value that passed reliably (3/3 runs) recorded; Fig. 2(b) summarises the result. Pure-NMPC (tight) is reliable up to 1.5 m/s, APF+NMPC (tight) up to 2.2 m/s, and APF+NMPC (wide) up to 2.81 m/s. Although the wide planner keeps far more clearance, it is the *fastest*: its larger look-ahead lets it begin avoiding early and turn gently. Tightening the APF to the same clearance as pure-NMPC removes this advantage and drops the maximum reliable speed to 2.2 m/s, only modestly above pure-NMPC. The APF speed advantage therefore comes from its wider look-ahead, not from the architecture: at matched clearance the two are close.

Look-ahead vs. clearance. A stronger test of the same claim is to hand pure-NMPC the wide planner's clearance and ask whether the speed advantage moves with it. We inflated the pure-NMPC keep-out until the minimum clearance became $\approx 2.3 m$; the commanded speed was then swept as before. Fig. 2 reports the outcome. Pure-NMPC (wide) is reliable only up to 0.5 m/s, more than five times below APF+NMPC (wide)'s 2.81 m/s at essentially the same clearance and also below pure-NMPC's own tight-clearance baseline of 1.5 m/s. The reason is visible in Fig. 2(a): the goal-only reference still points straight through the obstacle line, so a wider keep-out radius does not smooth the reference, it simply forces a larger lateral swerve at each

obstacle. At high cruise the tilt needed for that deviation grows past the point where thrust can still support the vehicle vertically, and runaway follows. Extra clearance therefore does not, by itself, unlock APF+NMPC (wide)'s speed on pure-NMPC: the missing ingredient is look-ahead in the reference generator, not headroom in the constraint. Indeed, once the margin is widened, each obstacle demands so large a swerve that pure-NMPC drops below even its own tight-clearance baseline. This is direct evidence for the central claim that the speed ceiling of the in-NMPC branch is set by reference smoothness, not by where the avoidance logic lives or how large a clearance margin it enforces.

Shared failure mode. Above this speed, every mode fails the same way (Fig. 3). As the lateral avoidance maneuver demands a larger tilt, the body thrust axis points increasingly sideways; once the attitude passes roughly 90° ($q_w < 0.7$) the thrust accelerates the vehicle horizontally instead of supporting it, producing a positive-feedback *attitude runaway* (speed spikes from 2 to 9 m/s and altitude collapses) that the 1 s horizon cannot recover. This dynamic limit, not the solver (which stays at 8.6 ms), sets the maximum reliable speed.

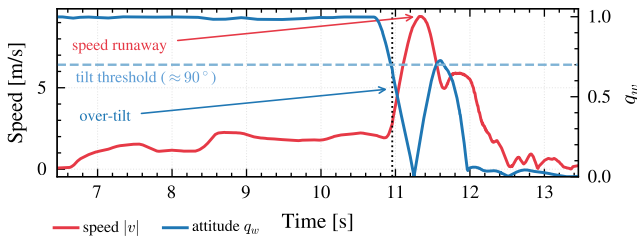


Fig. 3: Attitude runaway in a failed high-speed run. As q_w drops past the $\approx 90^\circ$ tilt threshold, the speed runs away and the vehicle crashes; the post-crash transient is cropped.

APF planner drawbacks. Delegating avoidance to an APF planner also has drawbacks that we observed. The potential-field parameters k_{att} , k_{rep} , d_0 and R_d are strongly coupled to the obstacle layout: parameter combinations that worked on a preliminary dense-world configuration produced chaotic loops or forced the drone to circumnavigate the entire obstacle line, and symmetric alternating obstacles amplify lateral oscillations, which is why the tight and wide configurations of Table I were tuned separately. The APF reference is also only kinematically collision-free and carries no dynamic-feasibility guarantee: the NMPC tracks it with error, and at high cruise speeds this error interacts with the attitude-runaway limit identified above. APF-in-the-loop is additionally known to trap the reference in local minima around U-shaped or concave obstacles; those geometries are not exercised by the in-line slalom studied here and are left to future work.

Attitude cost accuracy. The component-wise attitude cost introduced in Section II mildly underweights the error at large tilts relative to the geodesic form: starting from the small-angle equivalence, the relative penalty drops by 2.2% at 60° , 5.0% at 90° , and 18.9% at 180° . Across our tracking runs the 95th percentile of the attitude tracking angle stays

between 3° and 16° , where the underweighting is below 0.2%; only rare transient excursions extend beyond this band, and even in the worst case the underweighting is bounded by 20%.

Key metrics are summarised in Table II.

TABLE II: Obstacle-avoidance comparison (in-line slalom)

Mode	Avoidance by	Min. clr.	Reliable speed
pure-NMPC (tight)	NMPC keep-out	≈ 0.9 m	1.5 m/s
pure-NMPC (wide)	NMPC keep-out	≈ 2.3 m	0.5 m/s
APF+NMPC (tight)	APF planner	≈ 0.9 m	2.2 m/s
APF+NMPC (wide)	APF planner	≈ 2.3 m	2.81 m/s

Scope and outlook. These results come from a single in-line slalom in which every obstacle leaves a free corridor on either side, so the wide planner can trade space for speed. The advantage it gains from extra clearance is therefore tied to having room to keep it: in a denser or more cluttered scene the obstacles would not permit a ≈ 2.3 m margin, and the wide and tight configurations would converge. A fuller picture requires sweeping obstacle density, corridor width and layout, which is left to future work; the present study isolates *where* avoidance is computed at a single matched clearance rather than mapping the full envelope. The keep-out constraint (23) also treats each obstacle through a single centre and radius; extending it to more elaborate geometries is likewise left to future work: polyhedra can be enforced with one linear keep-out constraint per face [12], and non-convex or free-form obstacles could be handled through a signed-distance-function approximation [13].

B. NMPC Solver Performance

For the receding-horizon controller to run online, every optimal control problem must be solved within the 50 ms sampling period so that the control loop keeps pace with the planned rate. The SQP-RTI solve time was therefore profiled across the slalom scenario.

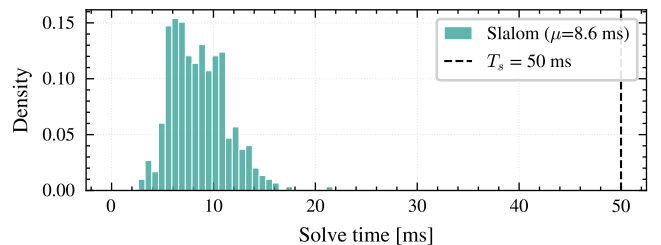


Fig. 4: NMPC solve time distribution for the slalom scenario. The dashed line marks the 50 ms sampling period T_s . All solves complete well below this budget.

The resulting distribution is shown in Fig. 4. The average solve time was 8.6 ms with a maximum of 21.4 ms across all samples, comfortably within the 50 ms budget, which confirms the real-time feasibility of the SQP-RTI approach. Table III summarises the solver statistics.

TABLE III: NMPC Solve Time Statistics (slalom scenario)

	Slalom
Mean	8.6 ms
Median	8.3 ms
Max	21.4 ms
< 50 ms	100%

V. CONCLUSION

This study presented a quaternion NMPC framework and used it to compare two places to perform obstacle avoidance: as keep-out constraints *inside* the NMPC, or in an APF planner whose references the NMPC tracks. On an in-line slalom at a matched ≈ 0.9 m clearance, in-NMPC avoidance was reliable to 1.5 m/s and the APF+NMPC tracker to 2.2 m/s; the larger gap to the wide-clearance APF (2.81 m/s) is explained by look-ahead, not architecture, since the APF influence distance couples clearance and path smoothness. This was verified directly by giving pure-NMPC the same wide clearance: reliable cruise dropped to 0.5 m/s, well below the tight-clearance baseline and more than five times slower than APF+NMPC (wide) at essentially matched clearance (Fig. 2), confirming that reference smoothness rather than constraint headroom is what sets the in-NMPC speed ceiling. Both architectures share an attitude-runaway speed limit set by the vehicle dynamics rather than the solver (8.6 ms average, well within the 50 ms budget). Used alone as a controller, an APF has no such dynamic guarantees [1], [2]; the APF-reference plus NMPC-tracker hybrid combines smooth, look-ahead planning with the NMPC's dynamic feasibility and is the most capable configuration tested. The quaternion state representation kept attitude tracking singularity-free throughout. Together these results suggest a simple design rule: when clearance is tight the in-NMPC keep-out alone suffices, whereas once clearance is available the speed advantage lies with a smoother reference, not a wider constraint. These findings come from a single corridor in which every obstacle leaves free space on either side; sweeping obstacle density, width and layout to map the full operating envelope, where the wide planner's clearance advantage is expected to shrink, remains future work.

REFERENCES

- [1] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *Int. J. Robot. Res.*, vol. 5, no. 1, pp. 90–98, 1986.
- [2] S. S. Ge and Y. J. Cui, "New potential functions for mobile robot path planning," *IEEE Trans. Robot. Autom.*, vol. 16, no. 5, pp. 615–620, 2000.
- [3] R. Verschuere, G. Frison, D. Kouzoupis *et al.*, "acados—a modular open-source framework for fast embedded optimal control," *Math. Program. Comput.*, vol. 14, no. 1, pp. 147–183, 2022.
- [4] G. Frison and M. Diehl, "HPIPM: a high-performance interior-point method for quadratic programming and model predictive control," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 6563–6569, 2020.
- [5] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *Proc. Eur. Control Conf. (ECC)*, 2019, pp. 3420–3431.
- [6] S. Sun, A. Romero, P. Foehn, E. Kaufmann, and D. Scaramuzza, "A comparative study of nonlinear MPC and differential-flatness-based control for quadrotor agile flight," *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3357–3373, 2022.

- [7] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Science Robotics*, vol. 6, no. 56, eabh1221, 2021.
- [8] M. Kazim, H. Sim, G. Shin, H. Hwang, and K.-K. K. Kim, "Aggressive trajectory tracking for nano quadrotors using embedded nonlinear model predictive control," *arXiv preprint arXiv:2312.01015*, 2023.
- [9] M. Kamel, J. Alonso-Mora, R. Siegwart, and J. Nieto, "Robust collision avoidance for multiple micro aerial vehicles using nonlinear model predictive control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2017, pp. 236–243.
- [10] B. Lindqvist, S. S. Mansouri, A.-a. Agha-mohammadi, and G. Nikolakopoulos, "Nonlinear MPC for collision avoidance and control of UAVs with dynamic obstacles," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 6001–6008, 2020.
- [11] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.
- [12] X. Zhang, A. Liniger, and F. Borrelli, "Optimization-based collision avoidance," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 3, pp. 972–983, 2021.
- [13] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, "CHOMP: Gradient optimization techniques for efficient motion planning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2009, pp. 489–494.
- [14] D. Falanga, P. Foehn, P. Lu, and D. Scaramuzza, "PAMPC: Perception-aware model predictive control for quadrotors," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2018, pp. 5200–5207.